

■ OFFENSIVE AI + CLOUD RED TEAMING ROADMAP 2026

The Ultimate Specialization Path for Penetration Testers

Built for You (Bug)

You already have strong pentesting foundations: Web/API, Exploits, Malware Dev, Endpoint Security, Reverse Engineering + basic AI knowledge.

Goal: Master [Offensive AI/LLM Red Teaming](#) + [Offensive Cloud Pentesting](#) and become a high-value hybrid Red Teamer in 2026.

- ✓ Step-by-step learning path (Free + Paid)
- ✓ Best YouTube channels & courses
- ✓ Hands-on projects & portfolio ideas
- ✓ Integration: Attack AI systems in the Cloud
- ✓ Certifications that actually matter

■ No fluff. Best of the best. Offensive focused. Red Teaming mindset.

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■ THE BIG PICTURE - YOUR ROADMAP

You already have a strong offensive foundation. We're layering two high-demand specializations on top: **Offensive AI/LLM Red Teaming** (your fastest path to differentiation) and **Offensive Cloud Pentesting** (for volume and modern relevance). The real power is in the **combination**.

PHASE 0: Foundation

Your current skills + basic AI/Cloud concepts (1 week)

PHASE 1: AI/LLM Offensive

Prompt injection → Agent attacks → Advanced red teaming (4-5 weeks)

PHASE 2: Cloud Offensive

IAM → Misconfigs → K8s/Serverless → CI/CD attacks (4-5 weeks)

PHASE 3: Integration

Attack cloud-hosted AI systems + combined red team scenarios (Ongoing)

Expected Outcome: In 3-4 months you'll be able to confidently red team modern AI systems running in cloud environments — a rare and extremely valuable skill set in 2026.

■ PHASE 1: OFFENSIVE AI / LLM RED TEAMING

This is your highest-leverage specialization right now. AI adoption is exploding, but very few people can actually attack these systems effectively. Your existing pentest + exploit skills transfer extremely well.

■ FREE FOUNDATIONS (Start Here - Week 1-2)

1. **OWASP LLM Top 10** — The bible. Study every category with examples.
2. **Microsoft AI Red Teaming 101** (YouTube full series) — Best free structured intro. Watch all episodes.
3. **Lakera Gandalf** — Gamified prompt injection training. Play until you beat all levels.
4. **MITRE ATLAS** — Adversarial tactics for AI systems. Map your attacks here.
5. **Promptfoo / Garak / DeepTeam** — Open-source red teaming frameworks. Install and experiment.

■ BEST PAID COURSES (Invest Here)

Course / Path	Provider	Why It's Worth It
LLM Red Teaming Path + AI-300 (OSAI)	OffSec	Most respected offensive security provider. Hands-on, practical attacks on real LLMs. Perfect for pentesters.
Certified Offensive AI Security Professional (COASP)	EC-Council	Tactical red teaming focus. Prompt injection, jailbreaks, agent attacks, guardrail bypass.
AI Red-Teaming Course	GTK Cyber	Practitioner-led, very hands-on. Strong on RAG poisoning and real-world scenarios.

■ MUST MASTER CONCEPTS

- Prompt Injection (Direct + Indirect) & Jailbreaking
- RAG Poisoning & Data Poisoning
- Model Extraction & Membership Inference
- Agent/Tool Abuse & Supply Chain Attacks on AI Pipelines
- Guardrail Bypass Techniques
- Multi-turn / Conversational Attacks
- Adversarial Examples for LLMs & Vision Models

■ TOP YOUTUBE CHANNELS - AI RED TEAMING

1. **Microsoft Developer** — AI Red Teaming 101 full series (watch this first)
2. **Antisyphon Training** — Practical OWASP LLM Top 10 red teaming sessions
3. **OWASP Foundation** — Official talks on LLM security
4. **Jarno Baselier (AI Red-Team Academy)** — Deep practical lessons on prompt attacks
5. **The Cyber Mentor / TCM Security** — General offensive + emerging AI content

Pro Tip: After watching Microsoft series, immediately start breaking real LLMs using tools like Garak or Promptfoo. Theory without practice is useless in red teaming.

■ PHASE 2: OFFENSIVE CLOUD PENTESTING

Cloud environments are the new attack surface. Most companies are moving to cloud, but very few pentesters know how to properly attack them. This is where volume of work + good pay lives in 2026.

■ FREE LABS & PLATFORMS (Week 1-2 Parallel)

1. **CloudGoat (AWS)** — Official vulnerable AWS environment by Rhino Security. Start here.
2. **FIAWS / FIAWS2** — Classic AWS attack labs. Learn metadata, IAM, S3 attacks.
3. **PwnedLabs** — Excellent free/paid tiers for AWS & Azure attack scenarios.
4. **Official Cloud Free Tiers** — AWS, Azure, GCP free accounts + intentionally vulnerable setups.
5. **Kubernetes Goat / Vulnerable K8s labs** — Container & orchestration attacks.

■ BEST PAID CLOUD OFFENSIVE TRAINING

Training	Provider	Focus
SEC588: Cloud Penetration Testing	SANS	The gold standard. IAM, misconfigs, container escapes, serverless, CI/CD. Very practical.
Offensive Cloud Foundations Path	OffSec	Strong on cloud-native attack vectors and CI/CD pipeline compromises.
GIAC Cloud Penetration Tester (GCPN)	GIAC	Certification path tied to SANS material. Good for resume.

■ CRITICAL CLOUD ATTACK SURFACES TO MASTER

1. **Identity & Access Management (IAM)** — Privilege escalation via roles, policies, trust relationships (biggest win area)
2. **Misconfigurations** — S3 buckets, security groups, public resources, metadata service abuse
3. **Container & Kubernetes** — Container escapes, RBAC misconfigs, pod security policies
4. **Serverless (Lambda, Functions)** — Event injection, permission boundaries, supply chain
5. **CI/CD Pipelines** — Pipeline compromise, secret exfiltration, poisoned builds
6. **Storage & Data Services** — S3, Blob, Cloud Storage attacks + snapshot abuse

■ TOP YOUTUBE FOR CLOUD OFFENSIVE

1. **Day Cyberwox** — Best dedicated cloud security channel (offensive + defensive)
2. **IppSec** — Masterclass in methodology (apply his style to cloud machines)
3. **HackerSploit** — Structured cloud + pentesting content
4. **Black Hills Information Security** — Excellent red team & cloud talks
5. **SANS Offensive Operations** — High-quality cloud attack demonstrations

■ PHASE 3: INTEGRATION — THE REAL POWER MOVE

This is where you become dangerous (in a good way). Most people do either AI or Cloud. Very few do both offensively. Companies are deploying AI **in the cloud**. Your ability to chain cloud IAM compromise → access to LLM endpoints → prompt injection / model extraction is extremely rare and valuable.

Example Attack Chains You Should Be Able to Build:

1. Compromise cloud IAM role → Access to SageMaker / Azure ML endpoint → Extract model or poison training data
2. Find exposed cloud storage with LLM training data → Poison dataset → Retrain attack
3. Compromise CI/CD pipeline that deploys LLM app → Inject backdoor into model or prompt
4. Serverless function hosting LLM agent → Abuse tool permissions → Lateral movement in cloud

Practice Environment: Deploy vulnerable LLM apps (using LangChain, LlamaIndex, etc.) on AWS/Azure free tier and attack the full stack.

Recommendation: After finishing Phase 1 and Phase 2 basics, dedicate 30-40% of your time to combined scenarios. This is what will get you noticed in interviews and bug bounties.

■ 12-WEEK ACTION PLAN (REALISTIC & AGGRESSIVE)

Week	Focus	Key Activities
1-2	AI Foundations	OWASP LLM Top 10 + Microsoft AI Red Teaming 101 series + Lakera Gandalf
3-4	AI Offensive Core	Start OffSec LLM Path or COASP. Daily tool practice (Garak, Promptfoo). Break 5+ LLMs.
5-6	Cloud Foundations	CloudGoat + FIAWS labs. IAM deep dive. PwnedLabs scenarios. Day Cyberwox videos.
7-8	Cloud Offensive	SANS SEC588 or OffSec Cloud Path. K8s + Serverless attacks. CI/CD focus.
9-10	Integration	Deploy LLM on cloud. Full-stack attacks. Chain IAM → LLM access → Prompt injection.
11-12	Portfolio + Polish	Document 4-5 full attack chains. Write reports. Update LinkedIn. Apply to roles.

Daily Habit: 1-2 hours minimum. Watch video → Immediate lab practice. Never watch more than 30-40 min without breaking something.

■ PROJECTS & PORTFOLIO BUILDERS

Theory is cheap. Red teaming jobs and clients want proof you can actually break things. Build these and document them professionally.

AI/LLM Projects:

1. Full OWASP LLM Top 10 report on a real/public LLM or app (with PoCs)
2. Custom red teaming framework/script for a specific use case (e.g., RAG app)
3. Multi-turn agent attack chain with tool abuse
4. Model extraction or data poisoning demonstration

Cloud Projects:

1. Complete CloudGoat / FIAWS walkthrough with methodology write-up
2. Kubernetes cluster compromise from external → cluster admin
3. CI/CD pipeline full compromise (build poison → secret exfil → lateral)

Integration (Gold Tier):

1. Deploy vulnerable LLM app on AWS (using free tier) → Compromise via IAM misconfig → Prompt injection / model attack
2. Full red team report on a cloud-hosted AI system (anonymized)

Portfolio Tip: Host write-ups on GitHub or personal site with clean formatting, screenshots, attack flow diagrams, and lessons learned. This beats any certification for serious roles.

■ CERTIFICATIONS WORTH YOUR TIME

Certs are signals. Choose ones that actually teach you something and are respected in offensive/cloud/AI spaces.

Cert	Area	Priority	Notes
OffSec AI-300 / OSAI	AI Offensive	★★★★★	Top choice for offensive AI. Practical.
EC-Council COASP	AI Offensive	★★★★■	Good tactical focus.
SANS SEC588 + GCPN	Cloud Offensive	★★★★★	Best cloud pentest training available.
OffSec PEN-300 / AWAE (if not done)	General Offensive	★★★★■	Strong web/app foundation if needed.

Strategy: Do 1 strong AI cert + 1 strong Cloud cert. Skip generic ones unless they directly help you get interviews in your target companies.

■ FINAL TIPS & MINDSET

1. Practice > Watching

Never spend more than 40% of your time watching. 60%+ must be breaking things in labs.

2. Document Everything

Every lab, every attack chain. This becomes your portfolio and interview material.

3. Join Communities

Discord/Telegram groups for OffSec, AI security researchers, cloud red teamers. Learn from others' write-ups.

4. Target Roles

- AI Red Teamer / Offensive AI Security Engineer
- Cloud Red Teamer / Cloud Penetration Tester
- Application Security Engineer (with AI/Cloud focus)
- Red Team Consultant (hybrid)

5. Mindset

You are not "learning AI security" or "learning cloud security". You are becoming someone who can **break modern AI systems running in cloud environments**. That's a rare and powerful skill in 2026.

You've got the foundation. Now go build something that makes recruiters and clients say 'We need this person.'

— Generated for Bug | July 2026 | Stay Offensive